Lecture 8: Linear Algebra Fundamentals, Eigendecomposition & SVD

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Agenda

- Part I: Vectors and Notation
- Part II: Matrix Operations
- Part III: Linear (In)dependence and Vector Spaces
- Part IV: The Determinant
- Part V: The Inverse, and Solving $\mathbf{Ax} = \mathbf{b}$
- Part VI: Eigenvalues and Eigenvectors
- Part VII: Singular Value Decomposition

This Lecture

There's an enormous quantity of information to cover on linear algebra, and I suspect many of you have already encountered it in some form. So today, we will:

- Take a guided tour through the relevant territory without stopping too long on any one concept
- For every concept, I will provide resources that cover it in detail from two sources:
  1. Khan Academy on Linear Algebra
  2. Savov's No Bullshit Guide to Linear Algebra (pdf on Moodle)

> ⚠ **This is the minimal working toolkit you'll require going forward**
>
> Please review any unfamiliar concepts independently using the suggested resources.

Part I: Vectors and Notation

Data Structures

1. Scalar: $x = 1$
2. Vector: a column of numbers

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

3. Matrix: a rectangular array, with generic element a_{ij} - row i, column j

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

More general a Tensor, elements $a_{ijk\ldots}$

Compact Notation

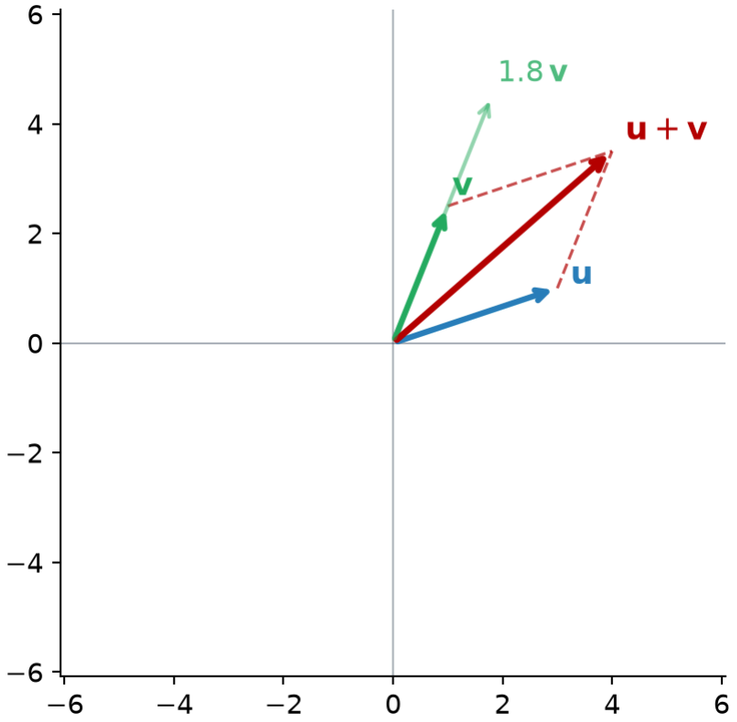
We will write the *shape* of an object rather than its contents:

$$x \in \mathbb{R} \qquad \mathbf{x} \in \mathbb{R}^n \qquad \mathbf{X} \in \mathbb{R}^{m \times n}$$

Read $\mathbf{X} \in \mathbb{R}^{m \times n}$ as "$\mathbf{X}$ is a real matrix with m rows and n columns".

A Vector Is a Direction and a Length

Two vectors add tip to tail; the sum is the diagonal of the parallelogram they span. Multiplying by a scalar stretches without turning.

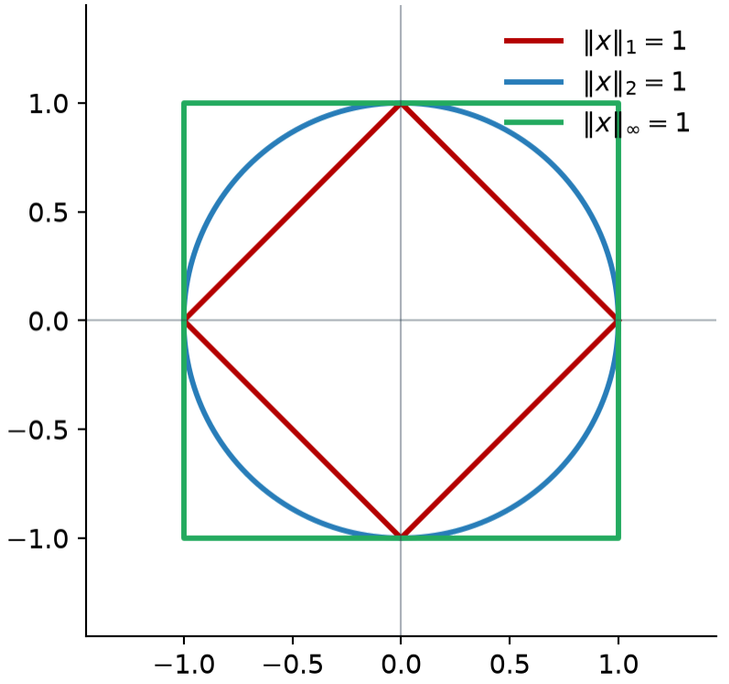


The Vector Norm

The norm of a vector measures its magnitude.

$$\|\mathbf{x}\|_1 = \sum_i |x_i| \qquad \|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2} \qquad \|\mathbf{x}\|_\infty = \max_i |x_i|$$

- The ℓ_2 norm is ordinary Euclidean distance and is the default. When we write $\|\mathbf{x}\|$ with no subscript, we mean it.
- All three are “vectors of length 1”, under three different ideas of length.



$\|\mathbf{x}\|_p = \left(\sum_i |x_i|^p\right)^{1/p}, \quad p \geq 1$

The Inner Product

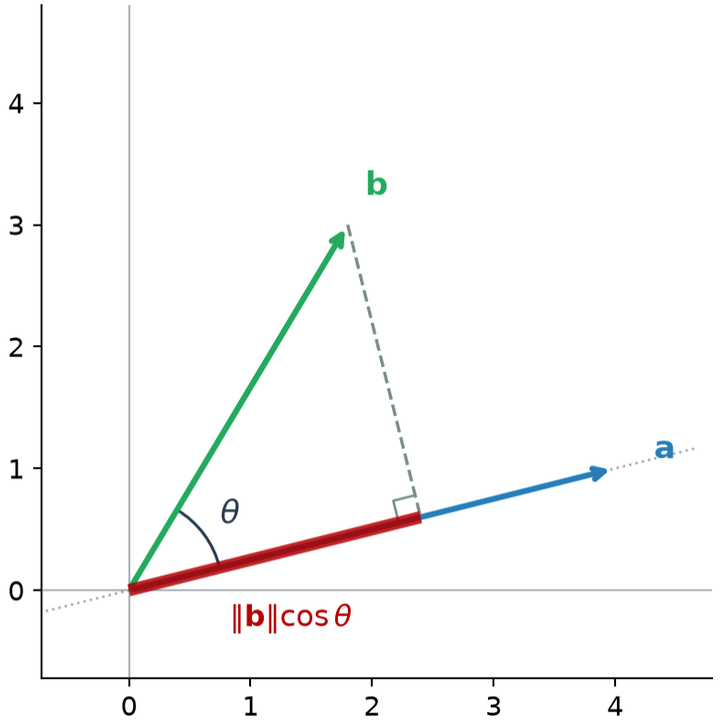
Definition. The inner product (or dot product) of $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is

$$\mathbf{a}^T \mathbf{b} = \sum_{i=1}^{n} a_i b_i$$

It is a scalar, and it connects to geometry through

$$\mathbf{a}^T \mathbf{b} = \|\mathbf{a}\| \, \|\mathbf{b}\| \cos\theta$$

So $\mathbf{a}^T \mathbf{b} = 0$ exactly when the two are orthogonal. And $\mathbf{a}^T \mathbf{a} = \|\mathbf{a}\|_2^2$ - the norm is an inner product in disguise.



Part II: Matrix Operations

Transpose

The transpose flips a matrix across its diagonal: $(\mathbf{A}^T)_{ij} = a_{ji}$.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \qquad \mathbf{A}^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}$$

An $m \times n$ matrix becomes $n \times m$. A column vector transposes into a row vector:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \mathbf{x}^T = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}$$

Transpose Facts

$$(\mathbf{A}^T)^T = \mathbf{A}$$

$$(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$$

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

$$\mathbf{a}^T \mathbf{b} = \mathbf{b}^T \mathbf{a}$$

Adding and Scaling

Addition is elementwise, and requires the same shape:

$$(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 4 \\ 5 & 6 & 6 \end{bmatrix}$$

Scalar multiplication is elementwise too:

$$(c\mathbf{A})_{ij} = c\, a_{ij}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \qquad c\mathbf{A} = \begin{bmatrix} c\, a_{11} & c\, a_{12} & c\, a_{13} \\ c\, a_{21} & c\, a_{22} & c\, a_{23} \end{bmatrix}$$

Multiplying Matrices

$$(\mathbf{AB})_{ij} = \sum_k a_{ik} b_{kj}$$

Entry (i, j) of the product is the inner product of row i of $\mathbf{A}$ with column j of $\mathbf{B}$.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} =$$

$$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{bmatrix}$$

To multiply, the matrices must be conformable:

$$\underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{B}}_{n \times p} = \underbrace{\mathbf{C}}_{m \times p}$$

The inner dimensions must match; the outer ones survive.

Consequently order matters: $\mathbf{AB} \neq \mathbf{BA}$ in general, and often only one of the two is even defined.

Systems of Equations

A system of linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

is exactly one matrix equation:

$$\mathbf{Ax} = \mathbf{b}$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

The Identity Matrix

$\mathbf{I}_n$ is $n \times n$ with ones on the main diagonal and zeros elsewhere:

$$\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

It is the multiplicative "1" of matrix algebra:

- it takes a vector to itself: $\mathbf{I}_n \mathbf{x} = \mathbf{x}$
- it takes a matrix to itself: $\mathbf{I}_m \mathbf{A}_{m \times n} = \mathbf{A}_{m \times n} = \mathbf{A}_{m \times n} \mathbf{I}_n$
- and it defines the inverse: $\mathbf{A}^{-1} \mathbf{A} = \mathbf{I}_n$

The Inverse - First Pass

Definition. The inverse of $\mathbf{A}$ is the matrix $\mathbf{A}^{-1}$ with

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n$$

- The inverse does not always exist; only square matrices may have an inverse (and sometimes they still don't)
- When it exists, several algorithms will find it
- It is mostly a theoretical tool and rarely needs to be computed directly

Part III: Linear (In)dependence and Vector Spaces

Linear Dependence

A set of vectors $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly dependent if at least one of them can be written as a combination of the others. I.e. it adds no new direction.

Definition. The set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is linearly dependent if and only if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$$

for some coefficients c_i that are not all zero.

Such a solution is called nontrivial. Setting every $c_i = 0$ always works; that is the trivial one, and it tells us nothing.

Three Examples

$$S_1 = \left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 6 \end{bmatrix} \right\}$$

The second is twice the first: $2\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$. Dependent.

$$S_2 = \left\{ \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right\}$$

No multiple of one gives the other. Independent.

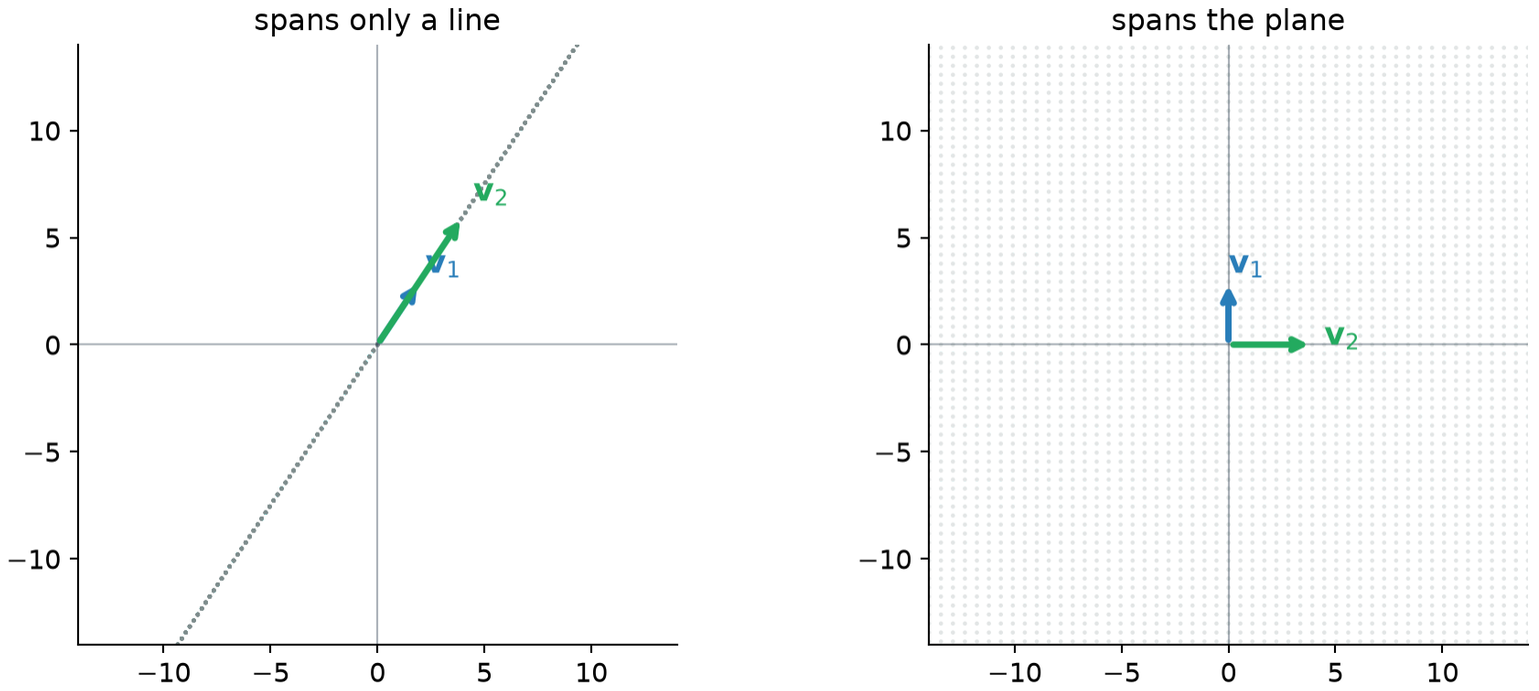
What about?

$$S_3 = \left\{ \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}$$

Dependent - and necessarily so. Why? Three vectors in $\mathbb{R}^2$ never can be independent.

The Span

Definition. The span of a set S is the set of *all* linear combinations of its members.



The dependent pair spans a line: the second vector adds no direction the first did not already have. The independent pair spans the whole plane.

Spanning Sets, Bases, Dimension

Going the other way, if V is a vector space and $\text{Span}(S) = V$, then S is a spanning set for V.

- A spanning set must have at least as many vectors as V has independent directions
- There are exactly 2 linearly independent vectors in $\mathbb{R}^2$, 3 in $\mathbb{R}^3$, and n in $\mathbb{R}^n$
- These independent vectors are said to form a basis for the space
- The number of vectors in a basis is the dimension of the space

What Is a Vector Space?

We have been using the term; here it is properly.

Definition. A vector space is a set V with addition and scalar multiplication, such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and scalars c, d:

- $\mathbf{u} + \mathbf{v} \in V$ and $c\mathbf{v} \in V$ (closure)
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutative) and $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associative)
- there is a zero vector $\mathbf{0}$ with $\mathbf{v} + \mathbf{0} = \mathbf{v}$
- every $\mathbf{v}$ has an inverse $-\mathbf{v}$ with $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
- $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$, $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$, $c(d\mathbf{v}) = (cd)\mathbf{v}$, $1\mathbf{v} = \mathbf{v}$

$\mathbb{R}^n$ is the example to keep in mind. But the axioms are deliberately weak, so that spaces of *functions* qualify too.

Part IV: The Determinant

The Determinant

The determinant of a square matrix - denoted as $\det(\mathbf{A})$ or $|\mathbf{A}|$, is a particular way to multiply the entries of the matrix to produce a single number.

We use determinants for all kinds of tasks:

- computing areas and volumes
- solving systems of equations
- checking linear independence
- deciding whether a matrix has an inverse

Computing It: 2×2 and 3×3

For a 2×2 matrix:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

For a 3×3 matrix, expand along the first row:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= a(ei - fh) - b(di - fg) + c(dh - eg)$$

Note the alternating signs. Each 2×2 determinant is what is left after deleting the row and column of its coefficient.

The General Formula for a Determinant

$$\det(\mathbf{A}) = \sum_{j=1}^{n} a_{ij}\, C_{ij} \qquad C_{ij} = (-1)^{i+j} M_{ij}$$

where M_{ij} is called the minor associated with a_{ij}, the determinant of the submatrix left after deleting row i and column j from matrix $\mathbf{A}$. The signed minor C_{ij} is the cofactor.

- Any row i gives the same answer; the 3×3 expansion in "Computing It" uses $i = 1$.
- The same works down any column j: $\det(\mathbf{A}) = \sum_{i=1}^{n} a_{ij}\, C_{ij}$.
- In practice, expand along the row or column with the most zeros - each zero entry drops a whole term.

$$\det\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

Geometric Interpretation of the Determinant

$\det(\mathbf{A})$ is a summary of the transformation that the matrix $\mathbf{A}$ imposes on an object.

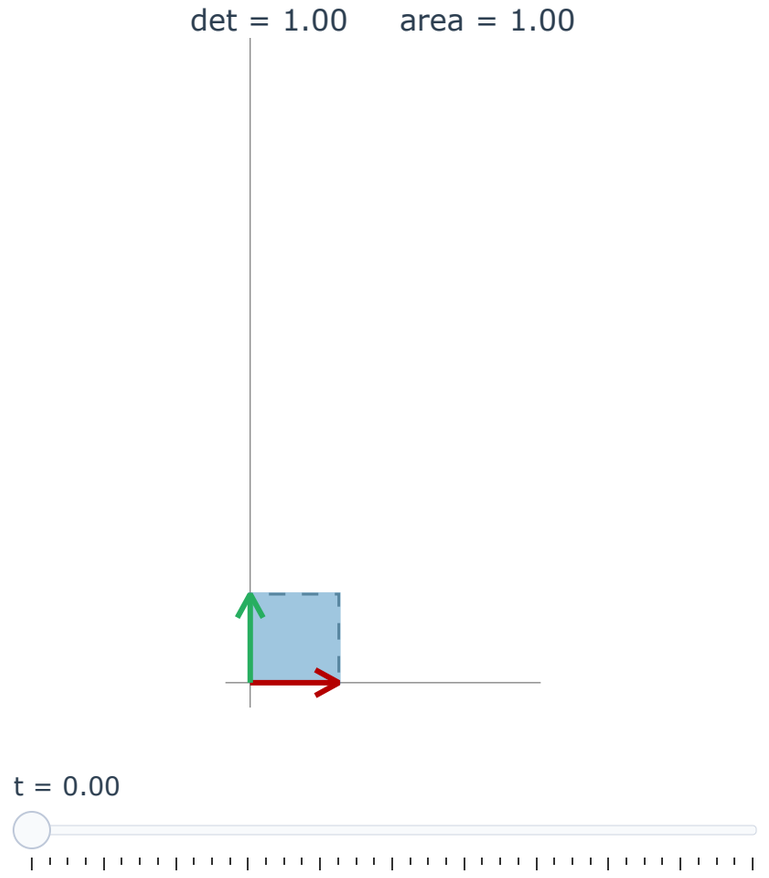
Scaling

$|\det(\mathbf{A})|$ is the factor by which area (in $\mathbb{R}^2$) or volume (in $\mathbb{R}^3$) is multiplied:

- $|\det(\mathbf{A})| > 1$ implies scaling up (expansion)
- $0 < |\det(\mathbf{A})| < 1$ implies scaling down (contraction)
- $|\det(\mathbf{A})| = 0$ collapses the dimensionality (e.g. an area becomes a line)

Orientation

The sign of the determinant indicates whether the transformation preserves orientation. Negative means a reflection.



Part V: The Inverse, and Solving $\mathbf{Ax} = \mathbf{b}$

When Does the Inverse Exist?

$\mathbf{A}^{-1}$ exists if and only if $\det(\mathbf{A}) \neq 0$.

And when it does, it is given by:

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \operatorname{adj}(\mathbf{A})$$

where $\operatorname{adj}(\mathbf{A})$ is the adjugate matrix.

The Adjugate matrix

$\mathbf{adj(A)}$ is defined as the transpose of the matrix of cofactors:

$$\text{adj}(\mathbf{A}) = \mathbf{C}^T \qquad C_{ij} = (-1)^{i+j} M_{ij}$$

For a 2×2 matrix this is:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

What's Really Important

You most likely never need to compute determinants or invert matrices by hand. What is most important to know is the following:

1. A matrix $\mathbf{A}$ has an inverse if and only if $\mathbf{A}$ has a nonzero determinant
2. A matrix $\mathbf{A}$ has an inverse if and only if its columns are linearly independent
3. Thus, a matrix has a nonzero determinant if and only if is columns are linearly independent columns

Three descriptions of one situation. We asserted the first claim previously and we will now prove the second claim.

A matrix is invertible if and only if its columns are linearly independent

We'll do a proof by contradiction: suppose a matrix with linearly dependent columns is invertible. Then we'll find an implication of this assumption that cannot be true - a contradiction.

Proof. Let $\mathbf{A}$ be a square matrix with linearly dependent columns $\mathbf{a}_1, \mathbf{a}_2, \dots \mathbf{a}_n$:

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

By the definition of linear dependence that means:

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \cdots + c_n\mathbf{a}_n = \mathbf{0} \qquad \text{with } c_i \neq 0 \text{ for some } i$$

A matrix is invertible if and only if its columns are linearly independent

That statement is a matrix equation:

$$\mathbf{Ac} = \mathbf{0}, \qquad \mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

If $\mathbf{A}$ was invertible, then we could multiply both sides by $\mathbf{A}^{-1}$:

$$\begin{aligned} \mathbf{A}^{-1}\mathbf{Ac} &= \mathbf{A}^{-1}\mathbf{0} \\ \mathbf{Ic} &= \mathbf{0} \\ \mathbf{c} &= \mathbf{0} \end{aligned}$$

But by the definition of linear dependence, $\mathbf{c}$ cannot be the zero vector. We have a contradiction. Hence, $\mathbf{A}$ must be linearly independent.

Part VI: Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

Definition. $\mathbf{v} \neq \mathbf{0}$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ (a scalar) if

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Multiplying by $\mathbf{A}$ usually moves a vector *and* turns it. An eigenvector is a direction that only gets scaled.

For example:

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\mathbf{v}} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \underbrace{5}_{\lambda} \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\mathbf{v}}$$

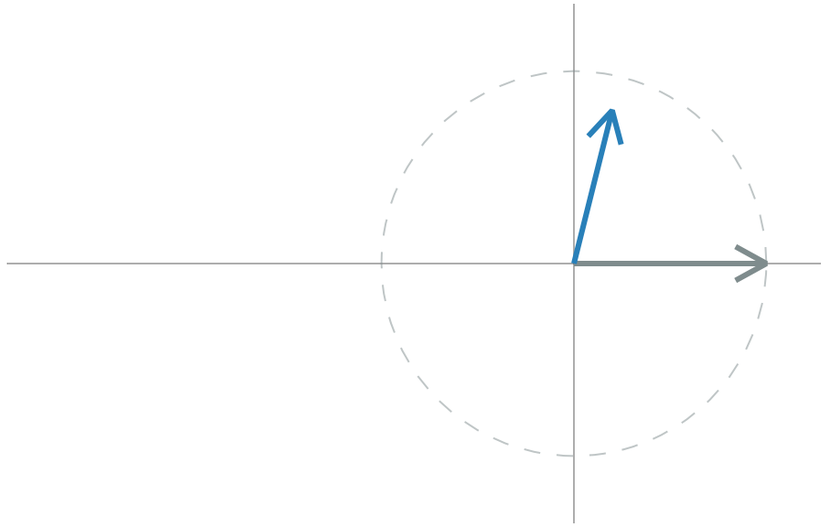
The Idea

Think of $\mathbf{A}$ as a transformation: it moves $\mathbf{v}$ from one place to another.

Most vectors get turned. A few do not: they come back pointing the same way, only longer or shorter.

If $\mathbf{A}$ is the journey, then λ is the essence of it. The part that survives when you strip out the turning. (*Eigen*: the self.)

Not every matrix can be decomposed in this way. But let's think about the properties of the matrices that can be.



Grey is $\mathbf{v}$, blue is $\mathbf{Av}$ (drawn as a direction). At most angles they disagree. At $(1, 2)^T$ and $(1, -1)^T$ - shown in red - they line up: $\mathbf{Av}$ is a multiple of $\mathbf{v}$.

The Condition for an Eigenvector

Start from the definition and rearrange:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

$$\mathbf{0} = \lambda\mathbf{v} - \mathbf{A}\mathbf{v}$$

Replace the scalar λ with $\lambda\mathbf{I}_n$, so both terms are matrices acting on $\mathbf{v}$:

$$\mathbf{0} = \lambda\mathbf{I}_n\mathbf{v} - \mathbf{A}\mathbf{v}$$

$$\mathbf{0} = (\lambda\mathbf{I}_n - \mathbf{A})\,\mathbf{v}$$

The Condition for an Eigenvector

$$\mathbf{0} = \underbrace{(\lambda \mathbf{I}_n - \mathbf{A})}_{\text{call it } \mathbf{B}} \mathbf{v}$$

Note that $\mathbf{v} = \mathbf{0}$ solves this trivially, but we required $\mathbf{v} \neq \mathbf{0}$, so we need a nontrivial solution.

This is precisely the definition of linear dependence for the columns of $\mathbf{B}$.

Furthermore, we know that if a matrix has linearly dependent columns then its determinant is zero.

In summary, if we want a nontrivial solution to the above, we require:

$$\det(\lambda \mathbf{I}_n - \mathbf{A}) = 0$$

Finding the Eigenvalues and Eigenvectors of a Matrix

Apply the condition to $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$:

$$\det\left(\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}\right) = \det \begin{bmatrix} \lambda - 1 & -2 \\ -4 & \lambda - 3 \end{bmatrix} = 0$$

$$(\lambda - 1)(\lambda - 3) - 8 = \lambda^2 - 4\lambda - 5 = 0$$

This is the characteristic polynomial. Factor it:

$$(\lambda - 5)(\lambda + 1) = 0 \qquad \Longrightarrow \qquad \lambda_1 = 5, \quad \lambda_2 = -1$$

The Eigenspace of $\mathbf{A}$

We have the eigenvalues. The eigenvectors come from putting each one back.

Definition. The eigenspace E_λ is the set of all $\mathbf{v}$ satisfying $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$.

For $\lambda_1 = 5$:

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 5 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \implies \begin{aligned} v_1 + 2v_2 &= 5v_1 \\ 4v_1 + 3v_2 &= 5v_2 \end{aligned} \implies v_2 = 2v_1$$

So $E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ - an eigenvector is never unique; any multiple of it is one too.

The second eigenvector is left as an exercise - repeat the eigenspace calculation with $\lambda = -1$.

Eigendecomposition

A square matrix with a full set of eigenvectors factors as

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$$

- $\mathbf{\Lambda}$ is diagonal, holding the eigenvalues
- $\mathbf{Q}$ has the corresponding eigenvectors as its columns, in the same order
- An $n \times n$ matrix has n eigenvalues - not necessarily distinct, not necessarily real

Read right to left: $\mathbf{Q}^{-1}$ rewrites a vector in the eigenvector basis, $\mathbf{\Lambda}$ scales each coordinate, $\mathbf{Q}$ translates back. In the right coordinates, $\mathbf{A}$ is just n separate stretches.

Example of Eigendecomposition

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\frac{1}{3}\begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}}_{\mathbf{Q}^{-1}}$$

with $\lambda_1 = 5, \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\lambda_2 = -1, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Symmetric Matrices

Definition. A square matrix $\mathbf{A}$ is symmetric if $\mathbf{A}^T = \mathbf{A}$.

Spectral theorem. A real symmetric matrix has real eigenvalues and orthonormal eigenvectors, so

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T, \qquad \mathbf{Q}^T\mathbf{Q} = \mathbf{I}$$

- The inverse comes for free: $\mathbf{Q}^{-1} = \mathbf{Q}^T$.
- The previous example is not symmetric, and its eigenvectors are not orthogonal: $\mathbf{v}_1^T\mathbf{v}_2 = (1)(-1) + (2)(1) = 1 \neq 0$.

  with $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

More generally, $\mathbf{A}$ is normal if $\mathbf{A}^T\mathbf{A} = \mathbf{A}\mathbf{A}^T$. The normal matrices are exactly those with an orthonormal eigenbasis, though it may be complex, as for a rotation. Symmetric matrices are the normal matrices whose eigenvalues are all real.

Why Bother: Powers

Suppose you need $\mathbf{A}^7$. Directly, that is seven matrix multiplications.

$$\begin{aligned}\mathbf{A}^7 &= \mathbf{AAAAAAA} \\ &= (\mathbf{Q\Lambda Q}^{-1})(\mathbf{Q\Lambda Q}^{-1})(\mathbf{Q\Lambda Q}^{-1})\cdots \\ &= \mathbf{Q\Lambda}\underbrace{\mathbf{Q}^{-1}\mathbf{Q}}_{\mathbf{I}}\mathbf{\Lambda}\underbrace{\mathbf{Q}^{-1}\mathbf{Q}}_{\mathbf{I}}\mathbf{\Lambda}\cdots\mathbf{Q}^{-1} \\ &= \mathbf{Q\Lambda}^7\mathbf{Q}^{-1}\end{aligned}$$

And a diagonal matrix powers up elementwise:

$$\begin{bmatrix}5 & 0\\ 0 & -1\end{bmatrix}^7 = \begin{bmatrix}5^7 & 0\\ 0 & (-1)^7\end{bmatrix} = \begin{bmatrix}78125 & 0\\ 0 & -1\end{bmatrix}$$

Further Resources for Eigen-everything

To review this material in greater detail, see Khan Academy Lesson 5: Eigen-everything, in particular:

- Introduction to eigenvalues and eigenvectors
- Proof of formula for determining eigenvalues
- Solving for the eigenvalues of a 2x2 matrix
- Finding eigenvectors and eigenspaces example

Part VII: Singular Value Decomposition

Singular Value Decomposition of a Matrix

Eigendecomposition only applies to square matrices! Why?

However a data matrix is $n \times p$: n observations, p variables. Almost never square.

The singular value decomposition does the same job - reducing a matrix to its essence - for a matrix of any shape:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \qquad \mathbf{A} \in \mathbb{R}^{m \times n}$$

The Three Matrices in SVD

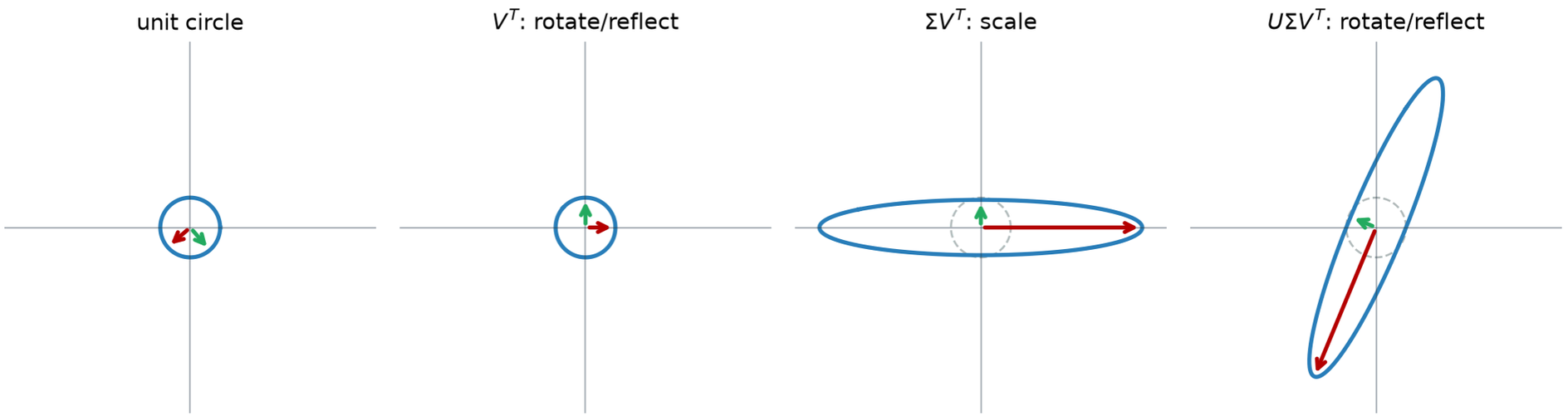
$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

- $\mathbf{U}$ ($m \times m$) holds the left singular vectors - the eigenvectors of $\mathbf{A}\mathbf{A}^T$
- $\mathbf{V}$ ($n \times n$) holds the right singular vectors as columns ($\mathbf{V}^T$ as rows) - the eigenvectors of $\mathbf{A}^T\mathbf{A}$
- $\mathbf{\Sigma}$ ($m \times n$) is zero except at positions $(1,1), (2,2), \ldots$, which hold the singular values $\sigma_1 \geq \sigma_2 \geq \ldots$ of $\mathbf{A}$ - the square roots of the eigenvalues of $\mathbf{A}^T\mathbf{A}$ (or $\mathbf{A}\mathbf{A}^T$)
- $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}\mathbf{A}^T$ have the same nonzero eigenvalues; the larger of the two fills up the rest with zeros

$\mathbf{A}\mathbf{A}^T$ and $\mathbf{A}^T\mathbf{A}$ are both square and symmetric whatever shape $\mathbf{A}$ has. That is the trick: we get back to eigendecomposition by squaring up.

$\mathbf{U}$ and $\mathbf{V}$ are orthogonal: $\mathbf{U}^T\mathbf{U} = \mathbf{I}$, $\mathbf{V}^T\mathbf{V} = \mathbf{I}$. This works because $\mathbf{A}\mathbf{A}^T$ and $\mathbf{A}^T\mathbf{A}$ are symmetric, and every real symmetric matrix is diagonalized by an orthogonal matrix (spectral theorem).

Geometrically: Rotate, Scale, Rotate



Every matrix does the same three things in sequence: rotate (or reflect) ($\mathbf{V}^T$), stretch along the axes ($\mathbf{\Sigma}$), rotate (or reflect) again ($\mathbf{U}$). The two arrows are the right singular vectors, carried along so the rotations are visible.

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

A Worked SVD

Take a genuinely rectangular matrix:

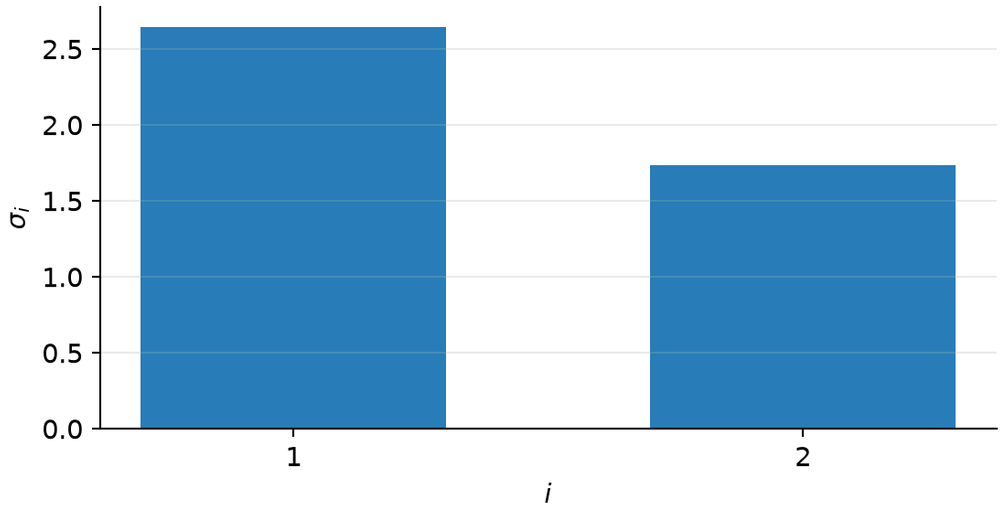
$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix} \in \mathbb{R}^{2\times 3}$$

Step 1. Square it up, the small way round:

$$\mathbf{A}\mathbf{A}^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 5 \end{bmatrix}$$

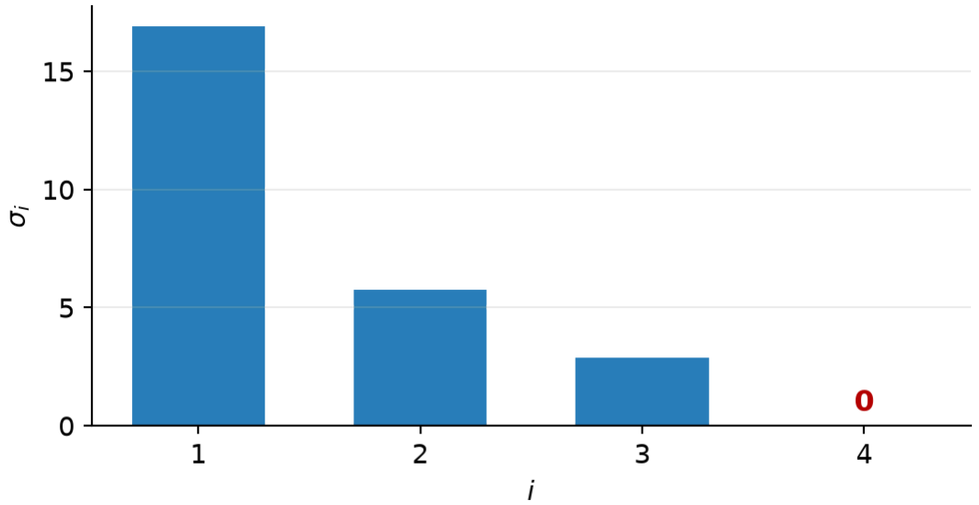
Step 2. Its eigenvalues: $\det \begin{bmatrix} \lambda - 5 & -2 \\ -2 & \lambda - 5 \end{bmatrix} = (\lambda - 5)^2 - 4 = 0$, so $\lambda = 7, 3$.

Singular Values Measure Structure



The worked example: $\sigma_1 = \sqrt{7} \approx 2.646$ and $\sigma_2 = \sqrt{3} \approx 1.732$. $\mathbf{A}$ stretches most along $\mathbf{v}_1$, less along $\mathbf{v}_2$ - and both directions carry something.

What If a Singular Value Is Zero?



Three directions carry the matrix; the fourth carries nothing. The number of nonzero singular values is the rank - the number of genuinely independent columns.

SVD Is an X-Ray of Your Data

- $\mathbf{U}$ and $\mathbf{V}^T$ describe how $\mathbf{A}$ rotates; $\mathbf{\Sigma}$ describes how it scales
- A large σ_i means a strong stretch: $\mathbf{A}$ maps $\mathbf{v}_i$ to $\sigma_i \mathbf{u}_i$
- A small σ_i contributes little; a zero one is pure redundancy
- If many singular values are near zero, the data can be represented in far fewer dimensions with almost no loss

Summary - linear algebra, eigendecomposition, and SVD

Concept	Key formula
Shapes	$\mathbf{x} \in \mathbb{R}^n, \mathbf{X} \in \mathbb{R}^{m \times n}$
Norms	$\|\mathbf{x}\|_1 = \sum_i \|x_i\|, \|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2}, \|\mathbf{x}\|_\infty = \max_i \|x_i\|$
Inner product	$\mathbf{a}^T\mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\| \cos\theta$; zero $\iff$ orthogonal
Transpose	$(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$ - the order reverses
Conformability	$(m \times n)(n \times p) = (m \times p)$
Linear system	$\mathbf{Ax} = \mathbf{b}$
Linear dependence	$c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$ with some $c_i \neq 0$
Determinant, 2×2	$ad - bc$; $\|\det\|$ = area scale, sign = orientation
Inverse, 2×2	$\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
The equivalence	invertible $\iff \det \neq 0 \iff$ independent columns
Eigen pair	$\mathbf{Av} = \lambda\mathbf{v}$
Characteristic polynomial	$\det(\lambda\mathbf{I} - \mathbf{A}) = 0$
Eigendecomposition	$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$, so $\mathbf{A}^k = \mathbf{Q}\mathbf{\Lambda}^k\mathbf{Q}^{-1}$
SVD	$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, any shape

Further Reading

Refresher

Khan Academy on Linear Algebra

Savov, *No Bullshit Guide to Linear Algebra* - Ch.2 Intro to Linear Algebra

Recommended

Savov, *No Bullshit Guide to Linear Algebra*

- Ch. 7.1, p. 257-268 (stop at Discussion) - eigenvalues and eigenvectors
- Ch. 7.6, p. 292-295 (stop at LU decomposition) - the singular value decomposition

Optional

- **Goodfellow, Bengio & Courville,** Deep Learning - linear algebra for machine learning Ch. 2, p. 29-43
- **Dan Kalman,** The SVD of a Matrix